

Fig. 3: An actual-size, single-side PCB for the automatic room power controller

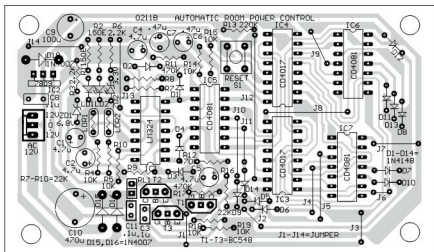


Fig. 4: Component layout for the PCB

blocked within this time, pin 14 of operational amplifier A4 becomes high making pins 5 and 6 of N2 high. This gives a clock pulse to pin 14 of IC4 via pin 4 of AND gate N2. However, as pins 8 and 7 of operational amplifiers A1 and A2 cannot become high at the same time, IC3 gets no clock pulse.

Thus when someone enters the

room, the output of IC3 advances by one. Similarly, when someone exits the room, the output of IC4 advances by one.

The identical output pins of decade counters IC3 and IC4 (Q0-Q9, Q1-Q1 and so on) are ANDed together via gates N3 through N12 and the gate outputs are connected together via

blocking diodes D5 through D14.

With this arrangement, if any two identical outputs of IC3 and IC4 are high, the output of any of AND gates N3 through N12 will be high. The high output of the AND gate makes transistor T1 conduct via resistor R16. As a result, transistor T2 cuts off and relay RL1 de-energises to deactivate the appliance connected to it.

Otherwise, transistor T2 conducts getting base current via resistor R18 and relay RL1 energises to switch on the appliance. At the same time, counters IC3 and IC4 reset via capacitor C3. These can also be reset manually by pressing 'Reset' switch S1 once.

An actual-size, single-side PCB for the automatic power controller (Fig. 1) is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

Possible modifications

The circuit can be modified to indicate the number of persons entering and exiting the room. For the purpose, you need to connect LEDs between the ground and the outputs of IC3 and IC4 through 1-kilo-ohm resistors.

One limitation of the circuit is that it cannot count more than ten persons in succession. When the eleventh person enters the room (without anyone exiting), the circuit resets. This problem can be overcome by cascading more decade counters. ●

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